

TIMARU, New Zealand

WMO: 937730

Lat: 44.3050S

Lon: 171.2214E

Elev: 27

StdP: 101.00

Time Zone: 12.00 (NZL)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-3.0	-2.1	-5.5	2.4	1.6	-4.4	2.6	1.7	10.3	9.8	8.8	10.8	1.9	310	0.544

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.7	26.7	17.0	24.6	16.4	22.6	15.8	18.8	23.8	17.9	22.4	17.0	21.0	5.3	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.0	12.2	20.7	16.0	11.4	19.8	15.2	10.8	18.7	53.5	23.9	50.6	22.4	47.8	21.1	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.9	8.2	6.7	DB	-5.0	32.3	1.2	2.7	-5.9	34.2	-6.6	35.8	-7.2	37.3	-8.1	39.2	(4)
			WB	-5.6	21.2	1.2	1.2	-6.5	22.0	-7.2	22.7	-7.9	23.3	-8.8	24.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.4	15.4	15.3	13.8	10.7	8.3	5.6	5.1	6.4	8.5	10.1	12.0	14.2	(6)
		DBStd	4.43	2.87	2.70	2.74	2.38	2.54	2.34	2.28	2.34	2.65	2.73	2.94	2.83	(7)
		HDD10.0	588	0	0	2	18	64	132	152	114	61	32	11	2	(8)
		HDD18.3	2905	99	91	143	229	310	381	409	370	296	255	192	131	(9)
		CDD10.0	749	169	149	120	39	13	2	1	3	15	36	70	133	(10)
		CDD18.3	24	9	7	3	0	0	0	0	0	0	0	1	3	(11)
		CDH23.3	326	117	73	39	3	1	0	0	0	1	8	27	57	(12)
(14)	Wind	WSAvg	2.6	2.9	2.7	2.5	2.3	2.3	2.3	2.2	2.3	2.6	3.0	3.0	2.9	(14)
		PrecAvg	601	59	53	53	51	48	41	44	45	38	52	55	61	(15)
		PrecMax	802	182	207	128	113	174	165	123	205	95	92	109	166	(16)
		PrecMin	453	5	14	8	14	7	2	5	7	8	16	24	14	(17)
		PrecStd	94	36	41	24	27	35	41	34	45	24	21	24	32	(18)
		DB	29.9	29.5	27.2	23.2	20.9	16.8	16.4	18.2	21.9	24.4	26.8	28.0		(19)
		MCWB	18.5	17.9	17.5	14.4	13.1	9.7	9.6	10.2	12.5	14.1	16.3	17.0		(20)
(22)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	26.6	25.3	23.9	19.6	17.4	13.9	13.5	15.3	18.8	21.1	23.2	24.7		(21)
		MCWB	17.3	17.2	15.9	13.6	11.6	9.2	8.4	9.6	11.6	13.2	14.6	16.7		(22)
		DB	23.9	22.8	21.5	17.4	14.9	11.9	11.4	13.1	16.4	18.6	20.5	22.0		(23)
		MCWB	16.9	17.0	15.6	13.0	10.9	8.3	7.6	8.8	10.5	12.3	14.3	15.8		(24)
		DB	21.3	20.9	19.3	15.7	13.1	10.3	9.7	11.2	14.3	16.1	18.1	20.0		(25)
		MCWB	16.3	16.2	14.9	12.4	10.4	7.7	7.0	8.0	9.8	11.5	13.4	15.2		(26)
		WB	20.6	19.9	19.1	15.8	14.2	11.2	10.6	11.4	13.8	15.6	18.0	19.4		(27)
(28)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	25.4	25.5	23.8	20.2	18.1	14.6	14.7	16.3	19.7	21.0	23.9	24.5		(28)
		WB	18.7	18.6	17.4	14.4	12.5	9.9	9.2	10.1	12.3	14.2	16.2	18.0		(29)
		MCDB	23.8	23.3	21.9	17.8	15.3	12.4	11.9	13.9	17.4	19.4	21.1	22.8		(30)
		WB	17.7	17.5	16.3	13.5	11.6	8.9	8.3	9.3	11.2	12.9	15.0	16.8		(31)
		MCDB	22.0	21.5	20.1	16.4	14.0	11.0	10.4	12.1	15.2	17.5	19.3	20.9		(32)
		WB	16.7	16.5	15.3	12.7	10.7	8.1	7.4	8.5	10.2	11.9	13.8	15.7		(33)
		MCDB	20.4	20.1	18.4	15.2	12.6	10.0	9.4	10.7	13.3	15.4	17.5	19.0		(34)
(35)	Mean Daily Temperature Range	MDBR	9.7	9.7	10.0	10.2	9.7	9.9	10.0	9.8	10.7	10.6	10.3	9.5		(35)
		5% DB	15.3	14.1	14.6	14.2	13.6	12.5	12.8	13.5	15.0	15.7	15.3	14.5		(36)
		MCWBR	7.6	7.4	7.9	8.9	9.0	8.7	8.7	8.8	8.9	8.8	8.2	7.6		(37)
		5% WB	12.4	11.9	12.3	12.1	11.0	10.5	11.0	11.3	13.2	13.3	12.9	12.2		(38)
		MCWBR	7.0	6.8	7.5	8.1	7.7	7.9	8.0	7.9	8.7	8.4	7.8	7.3		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.329	0.324	0.308	0.295	0.281	0.275	0.272	0.286	0.303	0.312	0.315	0.327		(40)
		taud	2.568	2.594	2.632	2.648	2.646	2.607	2.631	2.629	2.553	2.554	2.576	2.571		(41)
		Ebn at Noon	995	972	943	883	823	788	824	881	930	973	1003	1002		(42)
		Edh at Noon	103	95	83	69	57	53	56	68	87	98	101	104		(43)
(44)	All-Sky Solar Radiation	RadAvg	6.01	5.38	4.15	2.78	1.73	1.35	1.61	2.33	3.57	4.91	5.92	6.20		(44)
		RadStd	0.48	0.46	0.20	0.24	0.12	0.08	0.12	0.15	0.22	0.33	0.37	0.38		(45)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (5 neighbors)	+0.28	+0.28	+0.32	N/A	N/A	N/A	-41	N/A	N/A	(47)

Nomenclature: See separate page